

Business Problem questions

Funnel

1. What is the overall conversion rate?
2. Where is the largest customer drop-off?
3. What is the cart abandonment rate?
4. What is the checkout abandonment rate?
5. What is the payment failure rate?

Customer

1. What is the cart abandonment rate for new vs returning customers?
2. Which customer segment has the highest conversion?
3. Which customers generate the most revenue?

Product

1. Which categories have the highest abandonment?
2. Which products have high views but low purchases?
3. What is the conversion rate for sessions/orders with discounts vs without discounts?

Device

1. Is mobile conversion lower than desktop?
2. Which device has the highest abandonment?

Marketing

1. Which traffic source generates the most customers?
2. Which traffic source generates the highest conversion?

Delivery

1. How does conversion rate vary by estimated delivery time ?
2. Which warehouse has the highest late-delivery rate?

Payment

1. Which percentage of payment attempts failure rate?

Returns

1. Which category has the highest return rate?
2. What are the major return reasons?

Exploratory Data Analysis using python

Sales

- Total revenue
- Total orders
- Average order value
- Revenue trend
- Category revenue
- State revenue

Customers

- New vs returning
- Customer distribution
- Repeat purchases
- Customer revenue

Products

- Top 10 products
- Bottom 10 products
- High-view/low-purchase products

Funnel

Sessions



Cart



Checkout



Payment



Purchase

Delivery

- Average delivery time
- Late delivery %
- Warehouse performance

Returns

- Return %
- Return by category
- Return reasons

Deliverables

Data Preparation&Modeling (Python):

Clean and transform the raw dataset for analysis.

Data Analysis (SQL):

Organize the data into a structured format, simulate business transactions, and run queries to extract insights and Performed customer, product, delivery and return analysis to identify key drivers of conversion loss and generate data-driven business recommendations.

Visualization & Insights (Power BI):

Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.

Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.

GitHub Repository: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.